

Module 10.6: Response to Ethical AI

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IMT 574: Machine Learning

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November 30, 2025

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Artificial Intelligence is embedded into our everyday lives. However, there still needs to be a structured framework to establish a thorough understanding and explanation from Machine Learning (ML) and/or Artificial Intelligence (AI) systems to better promote transparency, responsibility, and accountability from both users and software providers. Imagine a world with AI usage but not knowing the accurate outcome, the users essentially are misinformed on the prompts' responses, and the providers continuing to apply the incorrect training into the software without governance or regulation. This is the sole reason why better transparency and explanation can benefit both users and organizations that help develop the programs. By doing this, users can confidently adopt the AI models and even vastly understand the system more effectively when putting it into usage. Next, this can also enable users to have the autonomy in making decisions and even deciding for their decisions, when necessary (for example: doctors can safely rely on an AI system (health diagnosis), to be able to form a decision and make that decision together with patients – promoting shared decision-making between stakeholders). Using from the same example between the doctors and the patients, the users (the healthcare providers) will understand which resources the information is from and even the logical reason behind this, given the fact that the users are presented with the full transparency in the AI software. Furthermore, users can also challenge (if necessary) any potential errors or bias, and they can even witness for themselves if the system operates fairly and without discrimination. According to Arrieta, et al. (2020), software providers can also gain advantage in the fair and trustworthy ML/AI system by gaining deeper insights into the models' internal workings, which then makes it easier to identify errors, debug issues, and even refine the models' performances and security systems over time. Also, explainable and transparent AI helps organizations have better data governance and regulatory policies that require certain audits and justifications during the decision-making processes, which really stir away from legal and reputation risks. Certainly, the important principles are trust and reliability, and transparency/explanation in ML/AI allows stakeholders to collaborate and to also seek / to receive feedback for better system efficiency and improvement (Lumenova, 2022).

In the context of “fair” system, Arrieta, et al. (2020) covers the following aspects: An ML/AI system needs to have possession of trustworthiness (when users can trust that the system will act as the way it is intended to, given any situations); causality (cause-and-effect relationships must be clearly defined – for example: AI predicts smoking links to lung cancer, but is that truly a true cause-and-effect relationship or simply correlation only?); transferability (AI should be able to reuse or transfer information from one context to another); informativeness (AI models are present to support human decision-making processes with excellent context and ability to avoid misinformation); confidence (can be related to ‘trustworthiness’ – ML/AI model must show stable and reliable knowledge when it is time to execute the command: (e.g.) An autonomous vehicle must be able to indicate whether an object ahead is a pedestrian or not based on interpretation and training); fairness (models are guaranteed to execute without any bias or discrimination); accessibility (models should always allow users to have access in decision-making or software development); interactivity (users to interact with the models by adjusting as necessary i.e. changing and testing variables to better understand the outcome); and privacy awareness (users' confidential information should not be leaked).

Next, we will continue using the example of healthcare provider using the ML/AI model to predict patients' risk scores. The system would be without fairness if the model is trained

completely on historical data, which contains some patients' test results missing or only certain groups of female patients could be subjected for certain lab tests, and that would leave out the rest of the ethnic or gender groups. As a result, the ML/AI model would be trained on such skewed data that the male patients may get lower risk scores just because the historical data did not show male patients would need many tests (Chinta et al., 2025). However, these male patients might exhibit much critical signs and would not get the appropriate treatments due to the incorrect risk score prediction. In contrast, patients would get better risk score if the system would execute the following: no bias involved (explainability tools reveal whether patients' sensitive attributes like race, gender, income status, etc. would be affecting the predictions just to be removed for better fairness); equal opportunity for all patients (patients should receive the correct risk scores regardless of (e.g.) ethnic group or gender); the ML/AI model certainly displays complete output that carefully explains why the patient's score is high with certain illness attributes; and the system must ensure patients' Health Insurance Portability and Accountability Act (HIPAA) is properly met, and patients' rights are completely protected.

Most importantly, concepts like bias, fairness, accountability, and transparency are related due to their existing relationships of one to another. The model contains bias, which shows the skewed and errors in the models, but this is when fairness needs to come into the picture to ensure any errors or bias would not happen across, and the usage would be fair to all stakeholders without promotion of discrimination or misinterpretation. With fairness, we then can connect with the organizations or the developers of the ML/AI models and require everyone to be accountable for how the systems operate. For example, there must be regulation, documentation, audits, or data governance to be completed and enforced for the sake of upholding such system. Without accountability, we would point fingers at each other without any further insights on how to resolve the problem. Next, transparency enters to enforce organizations ensuring the ML/AI models becoming not biased but transparent in their training and algorithm because organizations are being held accountable as well (Larock and Batorsky, 2021). As a result, this transparency also enables trust among stakeholders (organizations and users) in how they would see the models producing fair (and correct) outputs.

In an example where a system may be biased but fair, let's say we have a system that can predict patient's high-risk score because the hospital staff would like to prioritize the level of urgency to provide appropriate care for patients. Based on historical data, this facility determines (or has data overrepresenting) that older patients (age 65 or older) would likely exhibit more severe complications if they have experienced a fall (for example) versus younger patients (age 40 or younger). Hence, older patients in that age bracket have received much higher risk scores than younger patients. We can argue that the system is biased because younger patients with pre-conditioned injuries or disabilities may even be at the same or higher level of risk with the older patients. However, we all know that older patients tend to be at higher risk for injuries or severe complications based on consensus (Atinga et al., 2018), so the historical data is not biased as it does not reflect on patients' gender, ethnicities, or socioeconomic statuses. However, we can still argue that the system would be biased for age brackets. On the other hand, this model that the facility follows showcasing a genuine overview picture of older patients and their risks, so the hospital can appropriately prioritize the level of care urgency needed for each patient that would be admitted, which ensures patients' livelihoods safety and encourages hospital's proficiency.

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